2026–2027

ELEC0447 – Analysis of Electric Power and Energy Systems

Oral Examination Questions

*You will be assigned two questions, one from each list below. Questions from **List 1** are accompanied by a short numerical exercise (of the same type as those solved during the practical sessions TP 1 to TP 6).*

List 1: Modeling & Analytical Problems

1. **Three-Phase Systems and Power Transfer:**
   Describe three-phase systems, line-to-line vs. phase voltages, star and delta connections, and the per-phase equivalent circuit. Derive the active and reactive power transfer equations between two AC buses connected through an inductive branch.

2. **The Per-Unit Normalization System:**
   Explain the per-unit normalization principle, the choice of base quantities (S_b, V_b, I_b, Z_b), impedance transformation across voltage levels, and base change formulas. Discuss its usefulness and apply it to a multi-voltage example.

3. **Transmission Line Modeling:**
   Formulate the distributed parameter model (Telegrapher's equations), the characteristic impedance, and the propagation constant. Define the Surge Impedance Loading (SIL) and explain its physical significance. Derive the lumped π-equivalent model.

4. **Transformers and Voltage/Angle Regulators:**
   Describe the equivalent circuit of a real transformer, its per-unit model, and phase shifts in Y-Δ configurations. Explain the operating principle and power flow modeling of tap-changing (OLTC) and phase-shifting transformers (PST).

5. **Synchronous Generators:**
   State the steady-state per-phase model of a synchronous generator, draw its phasor diagram, and derive the power-angle equation. Describe excitation control ($E_{af} \sin\delta = \text{cst}$), reactances ($X''_d, X'_d, X_d$), and capability limits in power flow.

6. **Power Transfer Limits and the P-V Nose Curve:**
   Analytically derive the relationship between load active power P and receiving-end voltage V. Derive the maximum transmissible active power $P_{\max}$ and critical voltage V_{crit}. Explain the influence of power factor and line reactance.

List 2: Grid Operation, Control & Dynamics

1. **Power Flow Analysis:**
   Explain the power flow problem and bus classifications (Slack, PV, PQ). Derive the nodal mismatch equations and describe the Newton-Raphson solution algorithm. Contrast it with the DC power flow approximation, and explain single vs. distributed slack buses.

2. **Operational Security and Blackouts Analysis:**
   Explain the principle of the $N-1$ operational security criterion and how static security assessment is conducted. Choose and explain in detail one historical major grid disturbance or blackout (e.g., Tokyo 1987, Northeast US 2003, Italy 2003, Europe 2006, or Iberian Peninsula 2025): describe the initial events, the cascading mechanism, and the main technical lessons learned.

3. **Frequency Dynamics and Control Hierarchy:**
   Explain power balance and frequency evolution in an AC system (kinetic energy, inertia constant H, RoCoF). Describe the control hierarchy: primary (governor droop), secondary (AGC/aFRR), and tertiary control. Discuss the operational challenges associated with declining system inertia.

4. **Voltage Regulation and Voltage Instability:**
   Explain the physical mechanisms coupling active power to voltage angles and reactive power to voltage magnitudes. Describe the voltage control hierarchy (primary AVR, secondary, tertiary). Detail short-term vs. long-term voltage instability mechanisms, and discuss the impacts of high DER penetration (voltage rise, reverse power flows).

5. **Rotor Angle and Transient Stability:**
   Formulate the swing equation and describe every term. Explain the Equal Area Criterion (EAC) and how it relates to machine kinetic energy. Define the Critical Clearing Angle and Critical Clearing Time (CCT), and discuss the influence of damper windings, grid strength (SCR), and renewable energy integration.

6. **HVDC Transmission Systems:**
   Describe the main applications of HVDC systems and compare Line-Commutated Converters (LCC) with Voltage Source Converters (VSC). Detail the components of an LCC link, the operating principles of thyristor valves (firing angle α, extinction angle γ), control principles, and representation in power flow.

Write your answers and exercise development on separate sheets.
Indicate the date, your full name, and your student ID on each sheet.